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AI-Enhanced Design Thinking and Innovation Management: A Multifaceted Approach

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ABSTRACT

This study investigated the influence of AI on design thinking and innovation management of UK digital agency-based companies to enhance efficiency. However, this study employed a mixed-method study to understand the role of AI in design thinking processes and how it enhances innovation management. The quantitative study focused on exploring the influencing factors of innovation output (IO) in design thinking based on AI skillset and Expertise (ASE), level of AI integration (LAI), and implementation of AI technologies (IAI).
An online survey was conducted for participants (n=400) who are employees of UK-based digital agencies, such as CognitionX, Brainpool AI, Whoa, and The Data Lab. On the other hand, a semi-structured interview study was carried out for employees or managers (n=8) of companies.
The data findings from a quantitative study revealed that the level of AI integration has positively influenced the innovation output in the design thinking of companies. The qualitative study results depicted that these companies adopted AI tools successfully for training opportunities, design thinking, innovation management, and identification of trends and insights owing to the intuitive visualization and user-friendly features. Moreover, we identified challenges related to integration of AI, scalability, compatibility, and data privacy and security. These challenges could be alleviated by providing organizational support in terms of regular training, technical resources, cross-departmental collaboration, and personalized training about integrating AI into existing systems of design thinking procedures. Therefore, this study proved that AI tools enhance creativity and efficiency in design thinking and foster innovation in organizations.
Keywords: **Artificial Intelligence, Design Thinking, Innovation Management, Mixed-method Study, Efficiency.**

Dedication
I would like to express my deepest gratitude to my advisor, Dr. Sanjib Chakraborty, for his unwavering guidance and support throughout this research journey. His insights and expertise have been invaluable in shaping this study. I am also profoundly grateful to all the mentors who have provided their wisdom and encouragement, helping me navigate through various challenges. My heartfelt thanks go to my colleagues and professional friends, whose camaraderie and collaboration have been a constant source of motivation and inspiration. Without the support of these remarkable individuals, this thesis would not have been possible.

Acknowledgments
This thesis is dedicated to the memory of my father, who passed away. Dad, I did it. I hope you are proud of me. To my mother, whose love and sacrifices have been my foundation. To my beautiful wife, who stood by my side with unwavering support and patience. To my kids, whose encouragement has been my driving force. And to my brothers, who have been my pillars of strength. This work is a testament to the love, support, and inspiration you all have provided me.

INTRODUCTION

Research Background

In the global economy and the context of business, AI (Artificial Intelligence) stands as a thrilling prospect. Especially the concept of ML (machine learning) and AI lead to extend human performance or high match, practices on responsibilities, move the business's operational foundation fundamentally and also the practices of disrupt management obtain considerable possibilities (Krogh, 2018; Iansiti & Lakhani, 2020; Agrawal et al., 2017). AI leads to higher efficiencies, serves better outcomes or results, solves issues, advances or performs tasks, and improves human competencies (Wilson & Daugherty, 2018; Agrawal et al., 2019). AI stands as an emerging technology ruling certain game-changing services or products and concerting the current processes to conduct with more efficient quality. In a cheaper and faster form and the current

scenario, AI is accounted an effective general-purpose system (Brynjolfsson & McAfee, 2017). AI tends to move each sector, forming an expected 13 trillion GDP increase between today and 2030 as 100 years ago, the electricity conducted and 30 years ago, the internet conducted (Bughin et al., 2018).

According to Mendoza-Silva (2020), innovation capability indicates the competency of an organization or firm to employ and form new products, processes, or concepts conducting organizational value and its respective customers. In the current rapidly transforming business culture, innovation capability seems an essential component in ascertaining the success of an organization. "Engineering-based innovation" is considered an integrated term with innovation capability that indicates a systematic method of conducting emerging solutions, services and products depending on the techniques and principles of engineering (Sidhu, 2019). An efficient innovation capability empowers organizations to stay forward to competition, seize evolving opportunities, and respond to emerging obstacles. Organizations with AI innovation growth are competent to successfully employ AI technology, leading to an extremely competitive environment. Diverse businesses have enhanced their respective innovation competencies or capabilities in AI technology as an outcome of advancements, making them more profitable, productive, and effective (Kakatkar et al., 2020). Artificial intelligence has also developed enterprises' innovation capability significantly, permitting such enterprises to innovate and practice more effectively and faster than before. AI empowers firms to concentrate their respective assets on more strategic, creative and innovative tasks, including innovation and ideation, by reducing the effort and time needed to evaluate data and automating repetitive practices. Firms with AI tend to rapidly analyze and form broader ideas, conduct more informed decisions, and ascertain new opportunities or scopes. Therefore, this enables businesses or firms to establish more effectively and faster and introduce emerging services or products to market in the present dynamic business culture through driving competitiveness and growth (Igna & Venturini, 2023). Some concerns and arguments are also there despite several benefits carried by AI among people about AI's social and ethical responsibility (Hagendorff, 2020). However, social and moral responsibility contains the individual's and organizations' obligation or duties to perform appropriately that lead to enhances the greater welfare and also accounts for their actions' possible effect on others. Few concerns are there in the circumstances of AI that technology seeks to implement in several approaches that undermine human dignity and autonomy, perpetuate discrimination and bias, and violate privacy (Isaak & Hanna, 2018). For example, AI systems in lending and employment decisions are employed to perpetuate certain discrimination and prejudice. However, a concern is that AI is supposed to be implemented to displace workers and automate jobs, resulting in financial inequality and job loss. Thus, some people demonstrate that AI deployment and development are improper. The proper AI application in the contexts of law enforcement and military and questions regarding the AI systems' accountability and transparency (Felzmann et al., 2019).

Digital transformation is a major obstacle experiencing contemporary organizations or businesses (Jones et al., 2021). However, the requirement to support digital technology to adopt and establish emerging models of business forces organizations to reassess the present culture, structures, and capabilities to understand what technology or process is appropriate in business offerings and organizational processes, how they seek to be endorsed (Warner & Wager, 2019). Therefore, such profound transformations need organizations to check the old beliefs because they tend to establish policies that associate the relationship between harmful disruption and beneficial innovation. Finally, the literature reveals that digital transformation stands as a corporate business strategy's defining trait and executive management priority, organizations lead to obtaining a proper initiation on their respective transformation journey (Saarikko et al., 2020).

Digital technologies are becoming increasingly important in providing opportunities to access an unprecedented number of solutions. Frameworks for idea management and stages for crowdsourcing essentially support the generation and access of creative ideas. As a result, the world can be overflowing with ideas, forcing businesses that compete on display to remember that true value originates from many forms of innovation. The concept behind innovative solutions is to understand current problems more thoroughly. Innovations that are "more of the same" can be brought about by an underutilized solution, which can lead to gradual or drastic improvements. The process of coming up with a novel solution involves rethinking the problem by identifying an underutilized cause. Innovation revolves around underutilized methods, but especially underutilized explanations of why something is significant (Verganti et al., 2021).

In a particular context, design thinking indicates design phenomenology, including its objective (the novel solutions which it conducts, whether a process, service, or good; design decisions which are performed) and its process (through which collaborative practices, tools, methods, or phases; how to perform the design decisions). Therefore, design principles indicate the philosophy or insight that tends to inform the design performance and lead to introducing the design ontology. However, the differentiation between such two analysis levels empowers people to understand how AI might impact Design effectively. Thus, the scientific argument on design ontology is established in the design theories segment (Margolin & Buchanan, 1996; Margolin, 1989; Love, 2000; Galle, 2002). In organizational aspects, research study concentrates on design activities, obtaining a particular insight (such as design thinking). Hence, such insight supports the design theory research body and employs the concept to evaluate how in the business context, design-driven innovation can occur. However, the phrase "design thinking" possesses some uncertainty, the management research studies' efforts to focus its respective protocols towards three different significant components (Liedtka, 2015; Seidel & Fixson, 2013; Micheli et al., 2019; Dell'Era et al., 2020; Calabretta & Kleinsmann, 2017), such as:

- Abductive: To solve the issues, Design employs a creative method and fixes the problems in management aside from other practices of problem-solving. People represent the manager in management as fronting an alternative set from which a preference can be determined (Boland & Collopy, 2004). Therefore, such a decision attitude observes that practice with alternatives is easy to account for but complex to decide on among all. In contrast, toward problem-solving, the design attitude observes that forming an effective alternative is complex. Still, while an effective alternative is conducted, it won't be easy to make that decision regarding which alternative to determine. Thus, Design infers to assume the emerging one instead of searching for a particular solution. Whereas, Design is concentrated on the contingent and "how they might be" and not on the needs and "how things are" (Simon, 1982). It is obtained that Design led to solving issues via abductions from the logical inference insight instead of solely supporting "how things likely are" (inductive reasoning) and "how things are" (deductive reasoning), Design conducts via abductive reasoning (through conducting hypotheses regarding how things can be). For this reason, Design is integrated with ideation, innovation and creativity instead of analysis. As an illustration, research study can identify such principles in brainstorming activities.
- People-centered: While operated by Design, innovation was fostered by compassion with several users. Design-driven innovation comes from assessing an issue from the perception of the user or from conducting estimates regarding what is meaningful or helpful to that user instead of being operated by technology advancement. As an illustration, this research study can determine such principles and protocols in the ethnographic research segment.
- Iterative: The literature observes that abductions tend to be developed and employed continuously via advanced testing cycles. For learning and conversation, the prototypes conducted in such cycles perform as a particular playground. Therefore, in iterations, the prototypes involve both the users and the team, where solutions seem refined and tested. As an illustration, the research study seeks to determine such key protocols in conducting rudimentary mockup practices in the design process (Schrage, 1999).

Problem Statement

Notwithstanding that AI is employed to empower business processes by some particular enterprises and in such enterprises, AI innovations exist there. Still, the underlying reasons need to be clarified why these AI innovations tend to occur. However, previous literature mainly concentrates on illustrating a positive association, how AI systems led to an increase in the individual process of business, and how to implement AI systems to such innovation competency or capability (Reim et al., 2020; Kakatkar et al., 2020). However, such literature has yet to concentrate on why AI led to increased innovation competency. Thus, it is absurd regarding how AI affects an enterprise's innovation competency. The literature also does not focus on how AI led to fostering the elements possessing innovation capability. Moreover, some organizations determine the significance or requirements of developing their respective innovation competency in the market segment to stay competitive. In contrast, the literature assumes that AI technology stands as essential equipment to promote their respective innovation capabilities.

Organizations need to possess an appropriate assessment of how AI technology affects the organization's innovation capability. However, such innovation capability tends to be fostered by several factors belonging to diverse processes of engineering including product delivery, development, and research. The literature needs to focus on how AI affects such factors, which result in incorrect funds or resource allocation, finally leading to failure innovation. Previous research has initially concentrated on implementing traditional AI technologies' social and ethical values that seek to regulate dependent on certain pre-defined knowledge or rules. Therefore, AI technology possesses less uncertainty and can regulate easily leading to decreases in the harm risk to respective society and humans. In the research, notwithstanding the evaluation of generative AI's influence, an absence of concentration is there on how to secure its social and ethical responsibility. Some literature demonstrates that generative AI's ethics are specifically concerned with post-computation based on served products rather than evaluating the process (for example, Zhang, 2023). To confirm that such final products address the essential needs, it is significant to ensure the AI innovation capability's social and ethical responsibility. Hence, AI possesses the possibility to entirely convert the practices of innovation management by empowering a robust and efficient process of innovation. Finally, the idea of the AI implementation process still needs to be clarified for innovation management, and to obtain the proper method for AI implementation in organizational innovation efforts; the managers are still stressed.

Moreover, in organizational innovation's contemporary landscape, the AI and design thinking convergence caters for a convincing yet and stands as an intricate obstacle. Therefore, in this study, the intended issue is essential to assess the potential association between AI technologies and the processes of design thinking. Hence, the key issue falls in determining the way AI technologies' association fosters the essential tenets of ethical consideration, collaboration, impacting ideation, and design thinking.

Research Aim and Objectives

This study aims to comprehensively explore the multifaceted relationship between AI, design thinking, and innovation management. The study objectives are as follows:

- To investigate the impact of the degree of AI integration, adoption of AI, and skills and expertise in AI tools on innovative solutions in the design thinking process.
- To explore the user experience and adoption of AI in innovation management and design thinking.
- To determine and evaluate the limitations and challenges of AI systems in design thinking.
- To assess the organizational effect of AI adoption in innovation management and design thinking.

Research Questions

1. How do the level of AI adoption, AI integration, and skills and expertise in AI tools impact innovative solutions in the process of design thinking?
2. What factors affect the user experience and adoption of AI in innovation management and design thinking?
3. What are the primary limitations and challenges of AI systems in design thinking?
4. How does AI adoption affect organizational outcomes and processes in innovation management and design thinking?

Research Hypotheses

The below-mentioned research hypotheses were used to address the proposed research objectives:

H_{01}: Integration of AI has negatively influenced the innovation of solutions in design thinking within UK consultancies.

H_{a1}: Integration of AI has positively influenced the innovation of solutions in design thinking within UK consultancies.

H_{02}: The adoption of AI has negatively influenced the innovation of solutions in design thinking within UK consultancies.

H_{a2}: The adoption of AI has positively influenced the innovation of solutions in design thinking within UK consultancies.

H_{03}: Expertise and skills of AI have negatively influenced the innovation of solutions in design thinking within UK consultancies.

H_{a3}: Expertise and skills of AI have positively influenced the innovation of solutions in design thinking within UK consultancies.

Scope and Motivation

This particular study precisely explores AI adoption in the innovation management and design thinking segment. However, the research endeavours to lighten the involved challenges and practical implementation in AI integration within such segments. The study intends to disclose the relationship between design thinking processes and AI technologies. The scope of this research is surrounded to confirm specificity and depth. At the same time, the overviews that occurred are expected to significantly participate in AI-developed practices' wider assessment in the context of innovation management. Furthermore, the research also facilitates primary perspectives applicable throughout several organizational realms and leads to influencing a clear demonstration of certain symbiotic associations between innovation culture and AI.

The research has been influenced by certain changing landscapes constructed by AI adaptation in different organizational practices. Therefore, understanding the effective operations of AI and its subtle effect on innovation and design thinking is considered vital. Thus, the study seeks to emphasize the practical perspectives on the challenges and implementations of AI, participating and assisting in developing practices of innovation and informed decision-making. Moreover, the research intends to facilitate effective insights for firms employing the integration between innovation management, design thinking, and AI by investigating AI's transformative features in influencing problem-solving and creativity.

LITERATURE REVIEW

Design Thinking Paradigm

The management and business community has shown a real interest in Design Thinking, especially in the past decade. Its innovative thinking and responsiveness simultaneously made it one of the most widely used approaches to deal with problems that needed to be more solvable; however, corporate tools and procedures were unable to be handled for a number of factors (Baradaran, 2019). Owing to its flexibility and responsiveness during times of uncertainty, Design Thinking has become increasingly valuable as the world has grown more complicated. Another important factor contributing to Design Thinking's success is its comprehensive approach to the subject it was born into. This makes it especially appropriate for identifying and deciphering the functioning of complex systems, such as the advanced systems found in organizations. When faced with complexity, Design Thinking approaches issues by first taking into account both the issue in its whole and the larger context from which it originated. Design Thinking can consider problems from an original and unexpected point of view, rephrasing them in ways that make them solvable, according to this initial step back from the clamour of the matter

arena. The legitimacy of Design Thinking originates from the idea that dealing with complexity necessitates making decisions and assumptions—a process that some people do not seem to value. From a historical and pragmatic standpoint, the concept of Design Thinking quickly expanded and is now influencing an increasing number of fields in addition to the traditionally resistant ones (Baradaran, 2019). A review of the literature shows that there is no clarity in the ideas about what Design Thinking should be; instead, several various perspectives are presented.

Features of Design Thinking

The features of the design thinking process are examined in the following sections. Despite the fact that many explanations have been offered since the inception of Design Thinking, the following four features highlight the essential components (Baradaran, 2019):

- Prototyping: Design Thinking uses artefacts, illustrations, simulation, and other tools to create preliminary prototypes that lead to a testable structure (Lee & Park, 2021). Trial and error enhance decision-making and raise queries, but the primary goal of prototyping is to gather feedback from the final consumers. These prototypes offer quick and effective means for stakeholders to interact and gain knowledge from one another (Petrakis et al., 2021).
- Abductive reasoning: Plan Thinking combining natural thinking (knowing without thinking) and explanatory thinking (inductive and deductive coherent thinking that uses quantitative methodologies to arrive at findings) (Cramer-Petersen et al., 2019). The Design Considering ideation process helps to foster creativity by generating a large number of ideas that the group may then filter and distill down to the best, most sensible, or most creative ones (Micheli et al., 2019).
- Human-centered approach: Design Thinking incorporates human perspectives into all aspects of critical thinking. Typically, this involves observing the problem in its environment, coming up with ideas, conceiving, creating, and implementing a solution (Pata, 2021; Baker III & Moukhliss, 2020).
- Wicked problem: A wicked problem is defined as a situation that is difficult or phenomenal to unravel due to missing, conflicting, and shifting demands that are occasionally difficult to perceive (Earle & Leyva-de la Hiz, 2021). Design Thinking has become a problem-solving technique that may be applied to particularly unresolved issues (Baradaran, 2019).

Design Thinking Approaches

There are many objectives and obstacles that businesses must overcome during the development process. The author mapped four types of plans considering to encompass this advancement of the first perspective, following a detailed analysis of over 40 organizations: (i) Creative Problem Solving: comprehending difficult problems and incorporating both natural and analytical thought; (ii) Sprint Execution: communicating and testing tenable things to help move the solution forward and commit client information to memory; (iii) Creative Confidence: enables people to express themselves in innovative ways and become more certain; and (iv) Innovation of Meaning: conceiving of underutilized titles that suggest meaningful interactions for people (Verganti et al., 2021). Each of these plans will be covered in detail and explored in the sections below.

Design Thinking as Creative Problem Solving

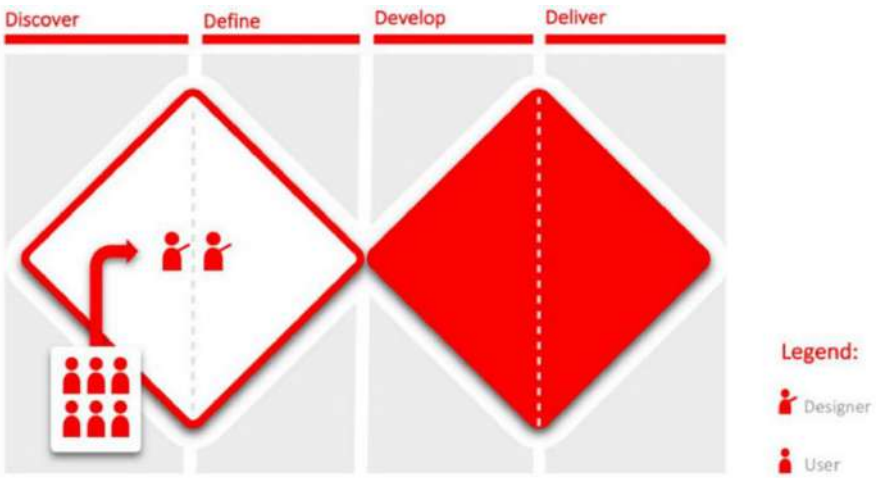
The most widely accepted and comprehensive explanation of Design Thinking is Creative Problem Solving. Convergent and divergent stages are typically substituted for and repeated during the creative process. During the divergent phase, a few ideas and propositions are made, and during the convergence phase, ideas and suggestions are honed and condensed to identify the most promising one. The design Thinking viewpoint advises applying these components for both defining the problem and developing the solution (Verganti et al., 2021).

The process is divided into four distinct stages where the first two deal with reframing the problem, and the last two focus on creating and presenting the right answer (Baradaran, 2019):

- Discover: This stage uses a variety of inquiry techniques to develop compassion with end users, uncover knowledge gaps, and highlight unfulfilled requirements. The process starts here as shown in Figure 1. Divergence is the energy at work, pushing the investigation of the contemporary scene to its farthest and most profound limits.
- Define: Interpreting and modifying customer requirements and organizational goals while drawing on the experiences gained from the previous phase has been critical. Together, the groups define the scope and the issue explanation, guiding the approach towards achievable, rational, and enticing outcomes.
- Develop: Design researchers generate theories that are rapidly prototyped and iteratively tested. Remote user testing is essential before approving concepts for the final solution. These tasks drive the approach toward the distinctive confirmation of the right answer.

- Delivery: In this last stage, the developed solution is improved, developed, and delivered to the market as a product, a service, or a framework. In addition to the methodology, the choice of unique benchmarks and refinements leads to creative problem-solving (Baradaran, 2019).

Figure 1: *Design Thinking as Creative Problem Solving*



Note. Adopted from Baradaran (2019).

Design Thinking as Sprint Execution

Once ideas are generated rather than carried out, clients are more interested in purchasing goods and services than they are in concepts when it comes to Design Thinking as Creative Problem Solving. The realization of the Design Thinking worldview, or Sprint Execution, is based on this assumption. The foundation of the Sprint Execution methodology is the Design Sprint (Verganti et al., 2021). One possible approach developed by Jake Knapp at Google Wanders is called Design Sprint. He describes it as a plan, prototype, and customer-testing approach comprised of a five-day handle for responding to fundamental trade inquiries (Koivumaa, 2017). Teams may be given a five-day, time-limited Design Sprint cycle to quickly complete well-defined goals and deliverables while gaining important insights. The approach strengthens development, facilitates user-centred thinking, realigns the team around a common goal, and expedites item shipment. Design Sprints help things move more swiftly towards significant outcomes and demand participation from all participants in the form of commitment, initiative, and teamwork (Verganti et al., 2021).

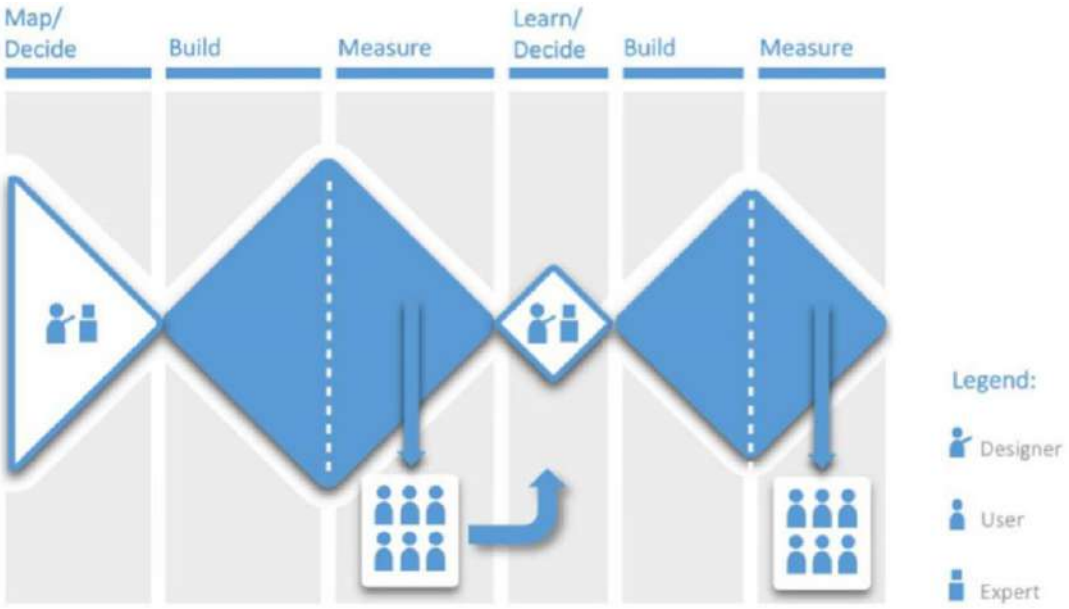
It is believed that Sprint Execution is the common progression of the Creative Problem-Solving methodology. Businesses can only thrive with continual improvement, yet the majority of them place a far greater emphasis on thought leadership than on actual execution. If ideation is viewed as glamorous and exciting, execution appears to be a tedious, untidy job done behind closed doors (Gavidarajan & Trimble, 2010). Either way, there is a growing significance associated with execution. Modern technological advancements that facilitate the translation of ideas into tangible products and accessibility, such as 3D printing and rapid prototyping tools, are made possible by the computer age.

Furthermore, ideas are being generated remotely through crowdsourcing phases to the extent that businesses no longer have to face this difficulty. The thought period is the path to pursue in situations were generating multiple ideas to satisfy customers' needs has become essential to survive. However, due to associated execution problems, this enormous number of ideas frequently does not result in appealing outcomes (Verganti et al., 2021). Sprint Execution maintains the iterative aspect of Creative Problem Solving. Still, instead of focusing on the generation of ideas, it places more attention on the downstream part of the process, with a notable emphasis on the development of products and services. The procedure is divided into four phases (Baradaran, 2019):

- Map/Decide: The method's starting point aligns with the Creative Problem-Solving approach's Discovery stage (Figure 2). However, in this instance, users do not initiate the process directly; rather, the readily available client-provided personal data ends up becoming the unique information source from which designers and experts (mainly product managers, social workers, and commerce designers) make decisions.
- Build: This stage is centered around making, with a focus on models that go beyond the Creative Problem-Solving methodology. The next step up for models is the least reasonable goods (MVPs) or offensive goods that users are required to value.

- Measure: As per the Incline Start-up technique, it serves as one of the fundamental rules of advancement bookkeeping. Teams using the Sprint Execution methodology need to be able to identify the appropriate metrics required for evaluating the performance of MVPs and then get feedback from users during the testing process (Ries, 2011).
- Learn: Understanding this step is essential for identifying whether the group's chosen direction is the right one. Every time the team gains knowledge, an idle cycle procedure that modifies the item based on feedback is activated. The aforementioned metrics then approve lessons (Baradaran, 2019).

Figure 2: Design Thinking as Sprint Execution



Note. Adopted from Baradaran (2019).

Design Thinking as Creative Confidence

One of the most significant problems to emerge in the last several years is creative confidence. The two Kelley brothers, Tom and David, 2013 introduced the idea of Creative Confidence. They gave them a variety of examples, techniques, and tips that gave them the confidence to stow away unutilized ideas. They exhibit Creative Confidence in the following ways: Fundamentally, creative confidence is essentially accepting of your ability to change the environment around you. It is the belief that you can easily complete the tasks you set out to complete. The core of innovation is this confidence in your ability to be creative and self-assured (Kelley & Kelley, 2013).

Establishments are facing novel and significant difficulties in securing and retaining their agents. First and foremost, people have been able to abuse exceptional entrepreneurial opportunities and technological breakthroughs that have clearly changed the existing economy thanks to advanced insurgency. Additionally, more commerce is needed to contribute to this. Fewer and fewer new businesses are making billion-dollar profits, and as a result, their importance to society is growing for both positive and pragmatic reasons. In addition, more people are emphasizing work-life balance or trying to find a personal calling and purpose in their profession. Since they are viewed as the two essential elements of success in this scenario, development and authority are becoming core values in every organization. As opposed to methods and tools, they are inherently connected to people (Verganti et al., 2021).

It is possible to link Design Thinking to the idea of changing the organizational culture, in addition to its original application in enhancing goods and services. The Creative Confidence technique is quickly taking off, utilizing the key components of Creative Problem Solving. Initiatives aimed at altering the organization's culture and mindset have made human-centeredness and empathy increasingly important. Intrapreneurs use top-down approaches to challenge established beliefs and presumptions, whereas entrepreneurs use bottom-up approaches to create organizational cultures and mentalities. Innovative intrapreneurs should foster a feeling of purpose among their staff members that drives productivity. Establishing a foundational internal structure is necessary to enable the development of feasible and efficient progression forms for Creative Confidence interior companies. Overcoming the fear of failure and presenting significant ideas to the showcase requires putting people's trust in themselves and encouraging them to reach their full creative potential. People are the true driver of innovation and the channel of change (Verganti et al., 2021).

It is possible to identify the following phases from a process perspective (Baradaran, 2019):

- Engage: It is crucial to secure important partners at this early stage, those who can spearhead the change. As a result, experts—who are typically internal partners—and designers gather as much information as they can on the flow that defines the business.
- Co-design: The most intriguing societal barriers preventing the organization from realizing its creative potential are determined by analyzing and mapping the data gathered during the first stage. The company's employees, who operate as designers and specialists, lock in to re-design the organizational display, reducing the traditional laziness towards change.
- Involve and Co-develop: The scale of the procedure increases in these two last stages. Larger segments of the organization—that is, the remaining employees—must be included to make the greatest noticeable organizational impact. The recognition of "small wins" or incremental actions that enable the effective implementation of change inside the organization's culture is crucial when managing change (Baradaran, 2019).

Figure 3: Design Thinking as Creative Confidence

Note. Adopted from Baradaaran (2019).

Design Thinking as Innovation of Meaning

The last emerging translation of Design Thinking that generally deviates from the mainstream perspective is the Innovation of Meaning. Innovation of Meaning originated from the need to find a solution to a need: making customers fall in love with a solution instead of having them enjoy it due to popular performances. Innovation of Meaning elevates creativity to a new plane by focusing attention on the "why" of problems rather than the "how." This kind of ingenuity creates something people will fall in love with and feel compelled to use. From a management perspective, this means repeatedly recommending something of value and providing a fully updated explanation of what is on display (Verganti et al., 2021).

Digital technologies possess a pivotal role in innovation management and AI-developed design thinking (Tigges et al., 2022). However, AI algorithms seek to evaluate a wide set of information to determine user behaviour, trends, and patterns, facilitating valuable overviews for the processes of design thinking. Therefore, AI empowers certain predictive modelling, assisting designers in anticipating the preferences and needs of users depending on traditional information, which leads to direct the phase of ideation (Pfau & Rimpp, 2021). Whereas AI-powered techniques seek to provide idea occurrence by evaluating information from several sources, proposing innovative ideas, and determining emerging trends. Digital technologies, such as AI-regulated collaboration platforms and communication tools develop knowledge-sharing and team interactions, influencing an environment of collaborative innovation (Descour et al., 2021). The technologies of AR and VR empower designers to test and conduct prototypes in several virtual environments, providing frequent refinement and iteration. Moreover, the association of digital technologies, such as cloud computing, blockchain, VR, AR, and AI, enables the process of innovation management and design thinking by facilitating advanced techniques for automation, prototyping, collaboration, and analysis (Savola et al., 2018).

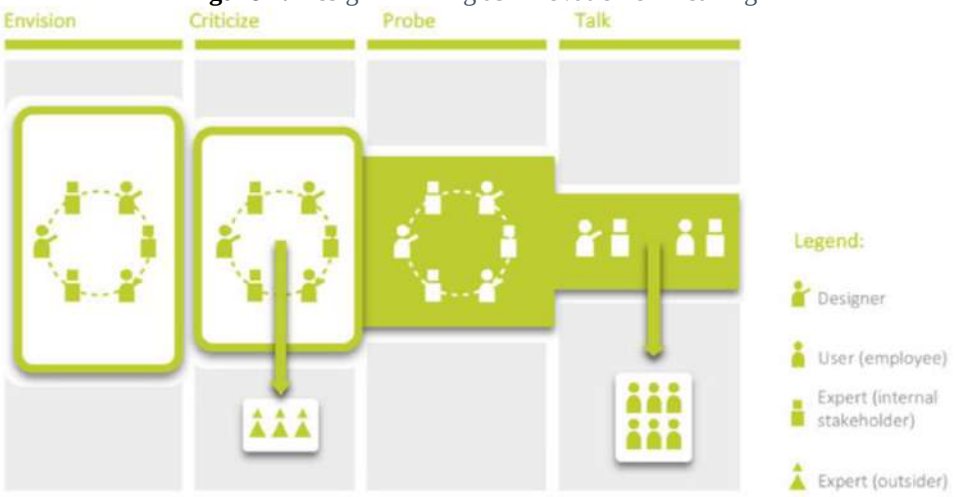
Craftsmanship ideation indicates to the generating creative process and establishing ideas or concepts with an effective focus on attention, quality, and skill to detail. Such craftsmanship ideation contains a meticulous method of idea refinement and generation, with emphasis on the craftsmanship's principles to confirm that the effective solutions are executed, well-crafted and innovative (Li et al., 2019). Craftsmanship ideation promotes the mastery of the skills in the ideas' execution and creation. However, the process concentrates on securing a high quality and precision level. Thus, ideas or concepts are refined via certain iterative processes, permitting continuous development (Manfredi et al., 2019). Whereas critiques assist

in revealing possible limitations and flaws in processes, strategies, or ideas permitting refinement (Cardiff et al., 2020). Controlling constructive criticism with several positive reinforcements seems significant to follow a healthy and effective culture of innovation (Cardiff et al., 2020). Innovation management confirms that innovation efforts are associated with the entire strategic organizational objectives and goals. Effective leadership support is an essential factor in influencing an innovation culture, encouraging risk-taking and facilitating resources (Guo et al., 2019). Often, innovation focuses on different aspects; cross-cultural teams provide diverse skills and perspectives. Innovation also influences a culture of adaptability and continuous learning, as innovation needs to stay with industry trends and new technologies (Iija, 2018).

In a world with endless options and no common goal, businesses fall victim to the idea catch-22: the more ideas they generate, the more they diverge from one another, and the slower progress is made. What matters to people also changes as a result of expansion in a world that is changing swiftly and consistently. By focusing on solutions, businesses often find themselves tackling issues that have ultimately shown to be pointless in the interim (Verganti, 2017). Creating new meanings for objects (products, services, business models, etc.) is a growing issue for traditional Design Thinking. Innovation of Meaning can be extremely varied and, in fact, the opposite of creative problem-solving. In the unlikely event that the Creative problem-solving technique operates in the other manner—that is, by walking outside, observing clients using the current product, and gathering insights from the study of such behaviours—the Innovation of Meaning approach operates in the same manner. It moves important things forward from the inside out, or, to put it another way, starting with ourselves. Furthermore, Innovation of Meaning necessitates the craftsmanship of feedback in a manner that differs unexpectedly from the Creative problem-solving approach, which is based on the craftsmanship of ideation. When our initial beliefs are challenged by feedback, they become stronger and more ingrained. Additionally, these two differences suggest important adjustments to the handling stages (Baradaran, 2019).

- Envision: Independently, internal personnel should produce the opening header. In other words, people focus on their personal beliefs and interpret what is important to them to build their own theory around "what they would adore people to love."
- Criticize: Considering how implicit the first stage is, the regulations are critical to avoid getting entangled in personal prejudices. The process should always be open to outside input in order to improve and energize individual headings. Feedback is first provided by coworkers who are in the same environment and culture, and then it is provided by outside professionals (collaborative players, specialists, and interpersonal translators).
- Probe and Talk: Designing assessments allows one to critique the material they cover and the ideas they represent. This is important both before and throughout this final phase of the process when end users provide their final assessment (Baradaran, 2019).

Figure 4: Design Thinking as Innovation of Meaning



Note. Adopted from Baradaran (2019).

Innovation Management

It is essential to grasp the fundamentals of innovation management, particularly in situations when innovation is essential to business expansion and competitive advantage. The series of actions taken to seize an opportunity and launch a concept is commonly referred to as the innovation process (Ahmed & Shepherd, 2010; Tidd & Bessant, 2020). Many methods for

managing innovation or more general R&D processes have been tried in the literature on innovation management; these methods range in stages from concept generation to product execution and commercialization (Rogers et al., 2014; Tidd & Bessant, 2020). Innovation consists of two processes: (1) coming up with an idea or invention and (2) turning that invention into a business or other practical use. Innovation encompasses all phases of the process, from the technical invention to the point of final commercialization. In order to more accurately depict the intricacy and diversity of tasks involved in the innovation process, a more detailed methodology was incorporated, segmenting the process into multiple stages that span from three to seven stages (Roberts, 2007). An example of an innovation process consists of four phases: (1) searching: looking for, evaluating, and managing signals about threats to and opportunities for innovation from the internal and external environment; (2) selecting: determining, in accordance with the innovation strategy, how the organization can respond to the signals; (3) implementing: pursuing pertinent ideas to develop new products and services; (4) learning: expanding the body of knowledge and continuously enhancing the innovation process through this cycle (Tidd & Bessant, 2020).

A creative organization will modify its process for innovation and specify the order and level of detail of its actions. Still, over time, different strategies for managing innovation have emerged, ranging from first and second-generation simple linear models to fourth and sixth-generation interactive models (Ahmed & Shepherd, 2010; Tidd & Bessant, 2020) that are progressively more complex. The innovation model's evolution is shown in Figure 5, and its phases—which are further defined by literature to help determine the course for future advancements in innovation management (Rothwell, 1994; Ahmed & Shepherd, 2010; Tidd & Bessant, 2020). The first generation, or "technology push," saw an increase in productivity across a range of sectors and industries in the 1950s as a result of new technological opportunities. Businesses placed a strong emphasis on R&D to enhance their offerings (Rothwell, 1994). The second-generation innovation model, also known as the market-pull model, was created as a result of growing competitive pressure and the realization that technology-push was failing in novel market contexts. The market-pull model brought the market perspective into the innovation process to overcome the technology-push model's blind spot for customers' needs, while the technology-push model placed more emphasis on research and development (Meissner & Kotsemir, 2016; Gaubinger et al., 2015). To cater to evolving customer needs, the majority of companies in this generation modified their current products (Nijssen & Mullick, 2023). This put businesses at risk of being overtaken by radical innovators and contributed to a further deterioration of research and development. In order to overcome these shortcomings, a push-pull innovation approach that combined market-pull and technological push was developed in the early 1970s, during the third generation of innovation (Mariani & Wamba, 2020; Beaudreau, 2023).

Figure 5: Evolution of Innovation Process Models

Generation	Model	Features
First – 1950s-	Technology push	Simple linear sequential model; emphasis on R&D
Second – mid-1960s-	Market pull	Simple linear sequential model; emphasis on marketing
Third – early-1970s-	Push-pull	Integration of R&D and marketing
Fourth – mid-1980s-	Parallel processing	Combinations of push and pull
Fifth – early-2000s-	E-integration	Integration of information technology (IT) into innovation systems
Sixth – mid-2000s-	Network innovation	Systems integration and extensive networking, continuous innovation

Note. Adopted from Mariani & Wamba (2020).

A sequential process and feedback loops were combined to create the innovation model, also referred to as the interactive "coupling" model (Rothwell, 1994). The need for rapid development to maintain competitiveness became evident over time as markets became more globally interconnected, competition grew, and product life cycles got shorter. The term "interactive-parallel processing innovation" was coined to describe the fourth generation, which emerged in the mid-1980s and was centered on integration and parallel development. While conventional innovation techniques, such as the well-known stage-gate model, were created for the development of new products in environments that were stable and predictable (Cooper, 1986; Smith, 2007), organizations must adapt to unexpected events when faced with change. Organizations are challenged to simultaneously explore new potential advantages and capitalize on current competitive advantages due to factors such as faster product life cycles, constantly changing customer needs, emerging technologies, and high levels of uncertainty (O'Reilly & Tushman, 2011).

As a result of criticism that traditional innovation processes were too rigid and linear to handle more creative and dynamic projects, it has been suggested in more recent research that the next generation of innovation processes should be more accelerated, flexible, adaptive, and agile (Cooper, 2014). This method uses several spirals or iteration loops to enable greater user experimentation and quicker learning. In addition to incorporating external networks during the initial stages of innovation, parallel integration and alignment of the various functional parties' activities can also enhance and expedite the innovation process instead of proceeding sequentially. The innovation model was further developed and integrated, and advancements in information technology prompted concurrent product development. Thus, in order to speed up the innovation process and increase flexibility, the fifth generation of innovation (2000), known as e-integrated innovation, concentrated on integrating IT-based tools. The sixth generation, also known as network innovation, emerged in the first decade of the twentieth century and is considered the most recent innovation model. It places more of the focus on networking and the strategic partnerships that facilitate both vertical and horizontal incorporation of third-party collaborators throughout the supply chain (Ahmed & Shepherd, 2010; Tidd & Bessant, 2020). To encourage working together with internal and external partners like educational institutions, research centres, businesses from various industries, and entrepreneurs as avenues for creativity and inspiration, open innovation approaches have been further embraced in the innovation management literature (Chesbrough, 2006). A variety of lean and agile innovation types have emerged to support innovation management agility and capitalize on start-up thinking, including innovation labs, accelerators, corporate incubators, jam sessions, and lean start-up camps (Kohler, 2016; Fecher et al., 2020).

Artificial Intelligence and Its Relation to Design Thinking

From a scientific perspective, the following section will concentrate on artificial intelligence (AI) and its relationship to human-centred Design. The first step in implementing AI for company operations in the present-day disruptive innovations-driven business environment is to evaluate the various journals' and academic papers' points of view regarding the significance of design thinking and human-centred Design (Baradaran, 2019). After that, how AI has changed over time, along with some definitions, will be discussed. In order to demonstrate the potential for natural intelligence in design ability, this is crucial. This section will then examine the perspectives from various scientific papers to determine whether the Design can be programmed. The function of AI across multiple designer tools will be discussed in the final section. This section's primary objective is to illustrate the value of design thinking for AI and vice versa. The second objective is to show the various studies conducted on this subject in scholarly reviews in order to identify any gaps in the field.

AI applications are reshaping businesses, though not all of them are doing so quickly. It is evident that AI now influences choices on everything from banking financial assistance to farming and that fully automated customer care was once unthinkable. AI is progressing through advances in technology and growing accessibility technologies, like data stores, powerful processing platforms, and development platforms. Currently, it is the perfect time for organizations to take advantage of AI (Baradaran, 2019). AI is predicted to boost the world economy by $13 trillion over the next ten years (Fountaine et al., 2019).

At present, machine learning (ML) and other forms of AI are being adopted by 80% of the biggest enterprises in their primary business. Five years prior, the figure was under 10 per cent. However, the majority of enterprises still employ AI tools as stand-alone programmes that need to be integrated into the larger IT infrastructure of the organization. According to the most recent evaluation of AI practices across over 8,300 major global businesses, it was found in what we believe to be the largest study on IT systems conducted by the enterprise to date (Ghosh, 2019).

AI is capable of more than just facilitating and modifying settings in accordance with personal preferences. AI has the potential to eliminate barriers related to culture and socioeconomic position in the future. It has become less problematic to use automatic translation in different languages. In a few years, learning environments will be able to connect students with teachers and tutors in accordance with their needs and interests, connect all students via contact lenses, and be integrated into immersive schools. According to Karuna Ammireddy's most recent Forbes post, AI remotely simplifies mental labour, just as physical tools and machines streamline physical tasks. These jobs supplement an already-existing workforce, yet they also call for the appropriate use of human labour and intelligence. He thought it would be just as disastrous to phase out employees in a server room full of AI software as it would be if a back office replaced its employees with unaccustomed laptop space. AI programmes are a tool to improve worker efficiency, not a replacement for human labour. Even so, many organizations' attempts are short-lived despite the promising prospects of AI (Baradaran, 2019). Only 8 per cent of businesses use fundamental methods that facilitate the broad adoption of AI and advanced analytics by numerous managers, according to data from the executive survey. Ad hoc pilots and one-phase AI use are the only approaches that most organizations focus on with AI (Fountaine et al., 2019).

Importance of human-centered Design in the Adaptation of AI in Today's Businesses

The upcoming era of AI developments has a lot of promise, but businesses need to figure out how and where to use it for specific problems across all departments. Furthermore, AI applications at present are only capable of limited operations. A

major mistake made by managers is to view AI as a plug-and-play technology that can be used immediately, as stated by (Fountaine et al., 2019). In order to launch some projects, they have started to spend billions on information infrastructure, intellectual property software tools, model development, and information knowledge. In organizational pockets, certain pilots manage to make a few dollars. The anticipated big winners, however, only materialize for a short time. Transitioning from pilots to company programmers can be challenging for businesses as they shift the emphasis from specific issues like bettering client segmentation to more general issues like optimizing client trips. Managers frequently take an overly limited view of what AI requires. In addition to skills and state-of-the-art technology, it is critical to align organizational culture, processes, and workflows to facilitate widespread AI adoption. However, the majority of non-digital businesses have conventional working procedures and viewpoints that are incompatible with AI. In order to set up the forthcoming era of technological advancements, C-Suite managers must take advantage of design thinking techniques to develop the middle manager funding and multidisciplinary cooperation required for company adoption. Ultimately, the significance of AI lies not in the working patterns of the technology but rather in the ability of companies to apply it (Baradaran, 2019).

This is a "transformation" that opposes the digital transformation rather than being "digital." How can one adapt, evolve, and create new things? The development processes used to develop new IT systems or technologies are not something we can rely on. Furthermore, conventional change management frameworks that emphasize creative thinking from the top down should be discussed. A new way of approaching work and thought is necessary for transformation. This views people as developers and participants rather than as recipients of change in organizations. And just that—wide-ranging dedication, inventiveness, and a commitment to change—is what design reflection offers (Verganti et al., 2021).

AI-based Innovation Management

A lot of tasks that were thought to be performed by humans require human cognition and can now be handled by AI-powered machines. Machines possess the ability to recognize intricate patterns, combine data, make inferences, forecast future events, and carry out tasks related to deductive reasoning. Hence, people are curious about how AI applications have an impact on innovation and make it possible for innovation management to become more efficient and successful (Brynjolfsson & McAfee, 2017; Agrawal et al., 2019).

Building on the innovation process, this study will explore and illustrate how AI can facilitate, expedite, or perform some tasks throughout the idea-to-launch process of innovation. There are four phases to the innovation process. The following sections will discuss some examples that AI could help with (1) identifying opportunities, such as monitoring promising innovations, brainstorming ideas, and determining customer requirements; (2) evaluating and selecting ideas, such as idea assessment and evaluation; (3) developing concepts and solutions, such as testing concepts and designing prototypes; and (4) launching the commercialization phase, which includes sales, marketing, pricing, etc (Rogers et al., 2014).

Opportunity Identification and Idea Generation Phase

When identifying opportunities and generating ideas, AI could assist with overcoming human limitations in information processing. Massive volumes of data are required for and processed by ML algorithms, which are used by AI systems, as reported (Brynjolfsson & McAfee, 2017). Their ability to identify issues, possibilities, and dangers exceeds regional search patterns and expertise areas, which could be advantageous in terms of finding and producing novel ideas. As it takes place, there are a number of fascinating uses for AI in a variety of industries (Haefner et al., 2021). In order to identify novel consumer requirements from nearly two million posts about body care, a German manufacturer of personal care products employed AI algorithms to analyze forum discussions online. Using information from their online postings, an American semiconductor chip manufacturer identified possible lead users and identified important issues related to their products using AI applications (Kakatkar et al., 2020). The examples demonstrate how AI can be a useful tool for generating ideas and opportunities, particularly when it comes to identifying pertinent customer problems and needs. AI offers more data to spark creative thought in at least two separate manners. Firstly, AI systems can provide insightful information while generating new ideas through analyzing solutions much more quickly due to sophisticated algorithms that use large amounts of training data and established rules, all facilitated by growing information-processing authority, as stated by (Haefner et al., 2021). Therefore, by resolving issues that were previously intractable, new insights are produced. This is particularly true for "structured" areas, including pharmaceutical, chemical science, materials research, and the biological sciences, where searching for answers used to be very time-consuming and costly since it required searching large combinatorial spaces. The most well-known instance of AI serving as the foundation for the development of innovative concepts in this manner is likely its solution to the "protein folding problem," one of the most important unsolved scientific mysteries (DeepMind, 2020).

AI can also be used to generate original ideas through more creative tasks, like making short films (Newitz, 2021), creative music (Macaulay, 2020a, b), and visual conceptions (Macaulay, 2020c). Art. Lebedev Studio, for instance, was developed

by Nikolay Ironov, a graphical design powered by AI. Prior to it becoming known that Nikolay was not a human, several businesses requested and financed Nikolay to develop unique visual identities for them, from simple logos to full brand identities. A neural network must be created on a text-image pairs dataset in order to produce original ideas. This training must be complemented by the application of additional algorithms that produce distinct fonts and colour schemes, as well as smooth, scale, and optimize the overall layout (Macaulay, 2020c). Nevertheless, there are still limitations and drawbacks, as coming up with new ideas is dependent more on dissecting preexisting solutions than it is on coming up with completely original ones. Idea generation powered by AI remains heavily reliant on the facts and data that are available (Füller et al., 2022).

Idea Evaluation and Selection Phase

Looking through a plethora of ideas during the idea selection phase calls for expertise and understanding. While having an experienced selection group and a well-defined set of assessment standards are essential for the process of choosing ideas, decisions tend to be made with incomplete or insufficient information, making them susceptible to cognitive biases and a lack of understanding. Additionally, selecting ideas has become exceptionally personal and frequently reliant on interpersonal relationships (Hofstetter et al., 2018). The opinion of the highest-paid individual is what drives the HiPPO method, which is still often used as the primary selection criterion (Lakhani, 2016). Processing data and making decisions can now be assisted by AI and, in certain situations, even taken over due to the development of ML algorithms (Verganti et al., 2020). Since AI can provide more unbiased knowledge and ideas, it may greatly assist the idea selection process. As suggested by the research study, it might enhance the company's capacity to identify and finance concepts that have the potential to generate its competitive edge in the near future. To help with deciding on the most promising ideas for products, an American food company, for instance, used unsupervised ML techniques (Haefner et al., 2021). (Kakatkar et al., 2020), they used an LDA topic model to analyze textual descriptions of solutions (such as flavour, colour, and taste). They then used a supervised random forest ML approach to determine which aspects of product ideas lead to high ratings. AI-based content analysis was used by innovation managers to select ideas with the help of this example, demonstrating how the introduction of AI may alter the way data has been dealt with (Füller et al., 2022).

Concept and Solution Development Phase

Building and developing solutions that are ready for market release is the goal of the concept and solution development phase. A novel good or service is created and evaluated through prototypes in order to gain knowledge about and optimize its viability, desirability, and feasibility, as well as move it towards being commercialized. As novel offerings have been developed, tested, and introduced, build-measure-learn loops are frequently used to gather feedback (Ries, 2011). During the concept and solution development phase, methods for collecting feedback and analyzing it are still crucial; however, AI offers a number of ways to accelerate learning and enhance these iterative cycles of experimentation. With the help of AI, data-driven decision-making and intuitive evaluation have been integrated into the product prototyping process. For instance, generative design apps (Krish, 2011) that rely on ML techniques have advantages in that they provide an extensive array of design alternatives and optimize manufacturing processes, materials, and expenses. To lower the weight of its automobiles, General Motors, for instance, used generative Design and investigated 150 design permutations. As a result, the parts are 20 per cent stronger and 40 per cent lighter than the initial parts (Autodesk, 2018).

Launch and Implementation Phase

Sales, transportation, shipment, and marketing to consumers tasks are all part of the launch and implementation phase. Businesses can use AI technologies and unique client data to provide organized goods and solutions (Kumar et al., 2019). AI has the potential to assist businesses in making real-time predictions about customer purchasing behaviour by leveraging their extensive customer base and transactions, which may include image, voice, text, face, etc. (Davenport et al., 2020). The only two companies that do this are Amazon and Netflix, which keep an eye on users' online activity and use recommendation algorithms to tailor the experience and suggest films or products to be purchased (Antonio, 2018; Füller et al., 2022).

Research Gap

The need for more research in the current body of literature pertains to the necessity of conducting a more focused investigation into AI-enhanced design thinking and innovation management. Although research on AI adoption is generally available, more information needs to be provided about the particular potential and difficulties that exist in this field (Ghosh, 2019; Kelley & Kelley, 2013). By providing a focused analysis of the dynamics surrounding AI integration, this study seeks to fill this knowledge gap and offer insightful advice for improving organizational tactics and encouraging creativity within the larger framework of design thinking (Verganti et al., 2021; Chesbrough, 2006).

METHODOLOGY

Methodology indicates the structured and systematic method applied to analyze, conduct, and plan research. This portion encompasses the rules, processes, and principles that leverage the data interpretation and collection to meet the research hypotheses and questions effectively. Therefore, the methodology selection is crucial as it ascertains the research findings' validity and reliability. It facilitates a roadmap for digital agency practitioners, confirming that their examination is well-organized, ethical, and rigorous. Thus, a well-established methodology increases the research credibility and permits others to build upon or replicate the work.

Research Design

The research followed an approach of mixed-method, merging quantitative and qualitative techniques of research. Qualitative data was collected via in-depth interviews with practitioners and industry experts from UK-based digital agencies, including CognitionX, Brainpool AI, Whoa, and The Data Lab, to acquire nuanced overviews of the practical challenges and applications of associating AI in innovation management and design thinking. Three in-depth interviews are conducted. Additionally, a quantitative survey is administered to collect structured data on the perceived effectiveness and impact of AI-enhanced practices. Questionnaires were provided to 400 participants who are employees of those enterprises and meet the eligibility criterion of a minimum of two years of professional experience. This is due to they possess expertise and practical insights into AI integration, contributing to a comprehensive understanding of design thinking and innovation management. Quantitative data is analyzed using descriptive statistics.

Qualitative data was analyzed using thematic coding and content analysis, identifying recurrent themes and rich narratives. With its ability to provide nuanced insights from industry experts and practitioners, qualitative data plays a crucial role in this mixed-method approach, enhancing the study. In-depth interviews give the depth and context necessary for a thorough understanding, exposing the real-world uses and difficulties of AI integration in design thinking. Contributing to the triangulation and validation of findings, thematic coding, and content analysis shed more light on recurring themes, adding rich narratives to the study. Integration of both quantitative and qualitative data sets offers a comprehensive understanding, allowing for triangulation and validation of findings.

For the qualitative study, a phenomenological or exploratory research design was employed, as this study aimed to assess the perspectives and lived experiences of practitioners and industry experts about the integration of AI in innovation management and design thinking. For the quantitative study, a cross-sectional and descriptive design was incorporated. However, a survey-based study assisted in gathering data on the perceived impact and effectiveness of AI-developed practices.

Research Approach

The inductive approach was adopted for the qualitative study, permitting patterns and themes to emerge organically from the data. The research study concentrated on UK-based digital agencies for an in-depth assessment by using a case study. On the contrary, the deductive approach was followed for the quantitative research, as it explores relationships and tests hypotheses by obtaining a "cross-sectional" survey. However, the study applied an approach of mixed-method containing the simultaneous analysis and collection of both quantitative and qualitative data. Therefore, qualitative data is collected with industry experts through in-depth interviews. Such a method facilitates a deep understanding of the practical challenges and applications of associating artificial intelligence in the context of innovation management and design thinking. Quantitative data is gathered via a survey among 400 participants, particularly the mentioned enterprises' employees, to measure the perceived effect and robustness of AI-empowered practices.

The approach of mixed-method permits for a more detail and comprehensive assessment of the research area. This approach merges both quantitative and qualitative methods' strengths, offering a holistic insight. Associating quantitative and qualitative datasets empowers the research findings' validation and triangulation. The approach confirms that the outcomes are reliable and robust by cross-verifying data from several methods and sources (Davidaviciene, 2018). Therefore, qualitative data acquired via an in-depth interview facilitates nuanced overviews from several industry experts. Such context and depth seem vital for a detailed assessment of the real-world challenges and uses of AI association in the realm of design thinking.

With industry experts, in-depth interviews facilitate the required context and depth for a thorough determination of AI integration's actual challenges and applications in both the segment of innovation management as well as design thinking. Content analysis and thematic coding of qualitative data facilitate to the validation and triangulation of study outcomes. Such methods determine rich narratives and recurrent themes, accumulating depth to the research. The amalgamation of both qualitative and quantitative data sets participates in a clear assessment of the study topic. Therefore, such a multifaceted method permits a more comprehensive and holistic investigation of AI-empowered innovation management and design thinking.

Data Collection Procedure

Several practitioners and industry experts in the qualitative analysis are selected from specifically UK-based digital agencies, such as CognitionX, Brainpool AI, Whoa, and The Data Lab. Hence, "purposive sampling" was adopted by this particular research to ensure the research participants possessed substantial perceptions and insights into the integration of artificial intelligence. A detailed interview was conducted both virtually as well as face-to-face. Several individuals were interviewed, and the interview duration lasted for approximately 60 to 90 minutes to obtain detailed perspectives. All the participants were chosen based on practical overviews and expertise in the adaptation of artificial intelligence. An initial contact was developed, and informed permission and consent was also acquired. Semi-structured interviews were developed, characterizing several open-ended questions to encourage participants to share their insights and experiences freely. The questionnaire was developed to cover several contexts of artificial intelligence adaptation in design thinking as well as innovation management.

A total of 400 participants were observed in the quantitative study from UK-based digital agencies, such as CognitionX, Brainpool AI, Whoa, and The Data Lab. Thus, a "stratified sampling" was selected, dependent on roles and departments. Therefore, a set of survey's structured questionnaire was conducted with the questions in the Likert Scale. However, the survey questionnaire was electronically distributed with a comprehensive explanation of instructions and purpose. The survey participants need to possess professional experience of at least two years to confirm a level of practical insights and expertise in the adaptation of artificial intelligence. The survey questionnaires were electronically distributed based on the participant's choices or preferences. Informed consent and clear instructions were facilitated with the particular survey. Hence, the survey contained certain structured questions concentrating on the perceived impact and robustness of AI-developed practices in the context of innovation management and also in design thinking. Multiple-choice questions and Likert scales are implemented to quantify the survey responses of the participants. A particular period or duration was allocated or fixed for the survey participants to fulfill the survey and timely reminders were also sent to the participants to increase the rates of response.

Data Analysis

An in-depth or detailed interview in the qualitative analysis was analyzed and conducted by content analysis and thematic coding. However, open coding is employed by particular axial coding to determine the patterns and themes. The software of NVivo was applied to organize the data systematically. Therefore, thematic coding contains systematically labeling and categorizing the topics, themes, and patterns determined in the qualitative research. Thus, each transcript of the interview was reviewed carefully, and a few initial coding was allocated to data representing segments relevant to concepts concerned with the association of artificial intelligence in both innovation practices and design thinking. Hence, codes were classified into wider themes, permitting the ascertainment of trends and patterns in the perspectives and experiences of participants.

Apart from that, the content analysis includes evaluating the textual data content systematically to derive meaningful and relevant overviews. The interview responses' content was assessed for noteworthy narratives, key phrases, and recurring patterns pertinent to the particular research questions (Pandey & Pandey, 2021; Gupta & Gupta, 2022). In data, the trends and patterns were determined, facilitating to an in-detail and comprehensive assessment of the practical obstacles and implementations of AI-empowered innovation practices and such design thinking. However, the responses' context was acknowledged to facilitate a nuanced data interpretation. Insights and themes derived from content analysis and thematic coding was synthesized to conduct an organized narrative that obtains the qualitative data essence. Illustrative examples and direct quotations from the conducted interviews that were involved to enrich and support the qualitative outcomes. Therefore, the qualitative outcomes were validated and compared against quantitative outcomes to confirm the comprehensiveness and consistency.

The survey responses in the quantitative analysis were evaluated by certain descriptive statistics (such as standard deviation, median, and mean). The software of SPSS was implemented for the analysis of quantitative data. Therefore, descriptive statistics, including standard deviation, percentages, frequencies, and means, were applied to describe and summarize the quantitative data's key features and characteristics. However, statistical measures were also implemented to determine the AI-empowered practices' perceived effectiveness or robustness in innovation management and also in design thinking. Thus, cross-tabulation was also applied to investigate the associations between diverse variables in particular survey responses.

Furthermore, the correlation analysis was developed to explore the direction and strength of associations between different variables, facilitating overviews of the possible integrations. Quantitative outcomes lead to triangulated and compared with qualitative outcomes to reinforce and validate the entire conclusions (Patel & Patel, 2019). Associating both data sets permits for a clear assessment, combining qualitative depth with quantitative precision.

Validity and Reliability

In the qualitative study, member checking of the interview participants maintained and applied, where each participant of the interview reviews the outcomes to confirm accuracy. Therefore, this particular research also ensured inter-coder reliability and validity by applying independent coding. The content validity in the quantitative study was secured by applying developed scales. Thus, the reliability of internal consistency was applied for questionnaire items implementing Cronbach's alpha. However, the internal validity indicates the level where the research appropriately computes the association between the intended variables to explore. A well-established plan of study with comprehensive casual integrations contributes to internal validity. However, the consistent methods of data collection among participants lead to an increase in internal validity (Mishra & Alok, 2022).

The external validity of the research understands the extent to which the research outcomes were generalized beyond the particular context and sample. The application of a representative and diverse sample, including employees from UK-based digital agencies, increases the external validity. The procedures and conditions of the research reflected the practical circumstances that contribute to the external validity. Comprehensively reporting the research participants and context permits the readers to understand the findings' applicability to other settings. The validity was constructed to understand whether such measures implemented in the research properly indicate the theoretical foundation under investigation. However, pilot testing the protocols of interview and survey assists in addressing and determining possible issues with the constructs' measurement. Utilizing validated and developed instruments and tools for several quantitative measures increases the validity of such a construct. They are implementing diverse methods containing quantitative and qualitative leads to strengthen construct validity and facilitate converging evidence. However, content validity seeks to explore whether the measure's content adequately indicates the complete interest domain. In the research field, finding input from professionals and experts confirms that the interview and survey questions' content is comprehensive and relevant. The method also ensures the research questions are associated directly with the study objectives and participate in content validity (Nayak & Singh, 2021).

Ethical Consideration

The participants in the quantitative study were completely informed regarding the rights, procedures, and purpose of the research. The research ensured participants' confidentiality and anonymity. Participants of the research had an absolute right to withdraw from the responses at a time, and they were fully voluntary. Whereas in the quantitative study, the purpose and confirms that all the participants are concerned and aware. To maintain the anonymity of the participants, the data was aggregated. Moreover, all data was protected and safeguarded to restrict unauthorized access.

Participants were entirely informed regarding the benefits, potential risks, procedures, and purpose of the research before obtaining consent from the participants. The research study facilitated a detailed and clear form of informed consent to all the participants, defining the voluntary or willful nature and characteristics of all participants and their respective right to withdraw without any consequences at any time. Protecting the participants' privacy by ensuring that their confidential information and data are kept personally, and without their explicit permission, such information will not be disclosed. Instead of several personal identifiers, the anonymized codes and data were used. The research study also securely stored all the data and restricted access to authorized team members. The study also confirmed that the participants' identities are not disclosed in any disseminated or published results.

Furthermore, this study assigned participants pseudonyms or codes in reporting findings to ensure anonymity. The study avoided any data that led to individuals' identification. The study also confirmed that the participants were voluntarily and willingly determined to participate without any such coercion.

In this particular study, approval or permission from an independent and institutional board of ethical review to confirm that the study addresses all the moral standards was acquired. The study developed comprehensive guidelines and specifications for authorship, considering the individual contributions involved in the study, such as collaborators and participants. All the responsible applications, integrity, and security of the collected data were confirmed. The study applied the secured practices of data storage, using data or information only for agreed-upon study purposes, and safeguarding against data breaches. Therefore, the study aims to support the research integrity, participate in ethical and responsible research practices, and protect the participants' rights by adhering to such ethical considerations in the realm of AI-developed design thinking and also in the segment of innovation management.

FINDINGS OR RESULTS

This section demonstrated the data findings to understand the impact levels of AI integration, skills and expertise of AI, and adoption of AI on improving innovation of design thinking processes while evaluating the challenges faced in implementing AI tools among UK digital agencies. However, we employed a mixed-method research design, including quantitative and qualitative studies. Using an online survey tool, quantitative research was conducted by considering employees from UK-based digital agencies as participants to determine the effects of AI adoption, integration, and expertise on innovation solutions in the design thinking process. The sample size of the study was 400, and quantitative analysis was carried out using IBM SPSS 26.0. On the other hand, qualitative research was included to evaluate the user experience of AI, challenges and limitations, and organizational effect of AI adoption by conducting a semi-structured interview for employees and managers (n=8) of UK-based digital agencies based on NVivo 14.0.

Quantitative Results

In this quantitative analysis, descriptive statistics, reliability analysis, correlation and regression analysis were performed through using the statistical IBM SPSS 26.0. This study included 400 participants to evaluate the impact of AI adoption, expertise, and degree of AI on innovation solutions. Descriptive analysis depicted the mean and standard deviation of data findings to derive the central tendency and variability of constructs, such as IAT ("Implementation of AI Technologies"), LAI ("Level of AI Integration"), ASE ("AI Skillset and Expertise"), and IO ("Innovation Output"). Reliability analysis was conducted for these constructs to find out whether the constructs maintain internal consistency and reliability. Moreover, correlation and multiple regression analyses were used to determine the influence of innovation solutions due to degree level, adoption, and expertise of AI in design thinking processes among UK-based digital agencies, such as CognitionX, Brainpool AI, Whoa, and The Data Lab.

Descriptive Statistics

Descriptive analysis was conducted on 400 participants' collected data to evaluate the general tendencies and variability of responses regarding the constructs of implementation of AI technologies (IAT), level of AI integration (LAI), AI skillset & expertise (ASE), and innovation output (IO).

Results depicted that the range was from 1 to 5 for all constructs, which means the participants used the entire scale to provide their opinions or choose their responses for the research questionnaire. Both IAT and LAI constructs had resulted in a mean of 3.7, indicating that participants showed a tendency towards implementation of AI technologies or tools and level of AI integration in innovation management. On the other hand, ASE and IO also had a higher mean score of 3.6 (Table 1). It recommended that participants had likely attitudes towards AI skillset and expertise and innovation output. When it comes to standard deviation, IAT showed a higher value of 1.1, which, represented there was more variability in the responses for the implementation of AI technologies. The variability of IAT was slightly higher than compared to LAI (1.0), ASE (0.9), and IO (0.9) (Table 1). This study reported that moderately to higher degrees of AI levels and AI adoption with some variability while consistency scores of AI expertise and skills and innovation output suggested that participants having similar experiences.

Table 1: Descriptive Analysis

Constructs	N	Minimum	Maximum	Mean (Std. Deviation)
Implementation of Technologies	400	1	5	3.7 (1.1)
Level of AI Integration	400	1	5	3.7 (1.0)
AI Skillset & Expertise	400	1	5	3.6 (0.9)
Innovation Output	400	1	5	3.6 (0.9)

Reliability Analysis

The reliability test was performed to determine whether the items in the scale maintained internal consistency. Cronbach's Alpha is an internal consistency predictor, suggesting how nearly similar an item's set as a particular group. However, it is acknowledged as a psychometric test's reliability estimation. The value of Cronbach's Alpha seems 0.961 directing extremely high reliability. A total of 32 items were adopted in the study to predict a similar underlying construct continuously.

From the obtained results, we could depict that the highest internal consistency resulted for the level of AI integration, i.e. 0.960 (Table 2). This was indicated as an extremely reliable construct to measure the attitudes of participants. Both IAT and IO showed an excellent internal consistency with 0.943 and 0.933, respectively. That means the items were very reliable in determining the implementation of AI technologies and innovation output. AI skillset & expertise also indicated good internal consistency but with lower scores compared to IAT, LAI, and IO. Still, it indicated a reliable measurement within the scale of AIE. Finally, this statistic presented that all the items were correlated highly with each other, strengthening the measurable tool's effectiveness or robustness.

Table 2: Reliability Analysis

Constructs	**Cronbach's Alpha**
Implementation of Technologies	0.943
Level of AI Integration	0.960
AI Skillset & Expertise	0.835
Innovation Output	0.933

Note. N=400 participants.

Relationship between Innovative Output (IO), Implementation of AI Technologies (IAT), Level of AI Integration (LAI), and AI Skillset and Expertise (ASE)

Pearson's correlation test was used to determine the relationship between all four constructs, such as innovation output, implementation of AI technologies, AI skillset & expertise, and level of AI integration.

Table 3 presents the correlation analysis showing significant positive associations between all variables' pairs. IO seems positively integrated with ASE, LAI, and IAT. However, the association stands comparatively weak as the Pearson correlation value appears to be 0.212, and the p-value is <0.001. Therefore, this indicated that other factors lead to possessing a substantial role while AI skillset, integration, and implementation participate in innovation. Whereas IAT possessed very effective associations with ASE, as the Pearson correlation value was 0.741 and the p-value was < 0.001, and LAI, as Pearson correlation value was 0.833, and the p-value was <0.001, directing that higher implementation of AI is closely integrated with higher AI expertise and integration.

Moreover, LAI is also effectively associated with ASE, emphasizing that firms with higher AI integration seek to possess more expertise in the field of AI because the Pearson correlation value seems to be 0.786 and the p-value is <0.001. Apart from that, a weak positive relationship is there between IO and LAI as the Pearson correlation value is 0.221, but the relationship is statistically significant as the p-value is <0.001. Whereas a weak positive relationship is there between IO and ASE as the Pearson correlation value is 0.185, the relationship is also statistically significant as the p-value is <0.001.
Such outcomes indicate that while the integration and incorporation of AI expertise and technologies possess a positive effect on innovative output, similarly, the associations are weak. Hence, the interconnection among AI integration, skillsets, and implementation seems strong, emphasizing the significance of a complete method to AI incorporation for increasing its advantages on innovation.

Table 3: *Correlation Analysis*

Correlations		IO	IAT	LAI	ASE
IO	Pearson Correlation	1	0.212**	0.221**	0.185**
	Sig. (2-tailed)		<0.001	<0.001	<0.001
	N	400	400	400	400
IAT	Pearson Correlation	0.212**	1	0.833**	0.741**
	Sig. (2-tailed)	<0.001		<0.001	<0.001
	N	400	400	400	400
LAI	Pearson Correlation	0.221**	0.833**	1	0.786**
	Sig. (2-tailed)	<0.001	<0.001		<0.001
	N	400	400	400	400
ASE	Pearson Correlation	0.185**	0.741**	0.786**	1
	Sig. (2-tailed)	<0.001	<0.001	<0.001	
	N	400	400	400	400

Note. p<0.01.

Influence of Implementation of AI Technologies (IAT), Level of AI Integration (LAI), and AI Skillset and Expertise (ASE) on Innovation Output (IO)

Multiple regression analysis was conducted to evaluate the impact of innovation output due to the factors of implementation of AI technologies, level of AI integration, and AI skillset & expertise.
In the model summary, the R-value stands at 0.221, the adjusted R Square is 0.044, and the standard error of estimations is 0.867 (Table 4). they are introducing the particular correlation coefficient, directing a weak positive association between the variables ASE, IAT, and LAI, and the dependent variable, such as IO. Apart from that, the R Square was 0.049, presenting that the variables describe an estimated 4.8% of the changes in IO. In the ANOVA test, the regression square sum seems to be 7.662, the residual square sum was 148.693, the sum of the total squares is 156.355, and the significance level, i.e., the p-value, is <0.01. The F-statistic seems to be 10.202, presenting that the entire regression framework is statistically significant. In the coefficient, it was observed that the IO's baseline levels while all the other variables, such as ASE, LAI, and IAT, were zero as the p-value stands <0.01 and the t-value is 11.560. LAI possess a positive and statistically significant effect on IO as the *p*-value is <0.01, whereas the IAT possess a positive but not statistically significant impact on IO as the p-value is 0.306 (Table 4).
Similarly, ASE also possesses a positive but not statistically significant effect on IO, as the p-value was 0.383. Therefore, this indicates that a statistically significant association was there between IO and ASE, IAT, and LAI. Moreover, this regression analysis reveals that the variables illustrate only a small portion of the changes in IO, while the entire model seems statistically significant.

Table 4: Regression Analysis

Model Summary					
R	R Square	Adjusted R Square	Std. Error of the Estimate		
0.221	0.049	0.044	0.867		
ANOVA					
	Sum of Squares	df	Mean Square	F	Sig.
Regression	7.662	1	7.662	10.202	<0.01
Residual	148.693	198	0.751		
Total	156.355	199			

Coefficients					
	Unstandardized Coefficients		Standardized Coefficients	t	Sig.
	B	Std. Error	Beta		
(Constant)	2.867	0.248		11.560	<0.01
LAI	0.206**	0.065	0.221	3.194	<0.01
IAT	0.092	0.730	0.466	0.052	0.306
ASE	0.028	0.252	0.802	0.018	0.383

Note. p<0.01.

In summary, quantitative data findings showed that the level of AI integration, implementation of AI technologies, and AI skills and expertise were correlated with the innovation output based on correlation outcomes. Regression analysis revealed that the level of AI integration significantly influenced the innovation output, while AI skills and expertise and implementation of AI technologies did not impact the innovation output.

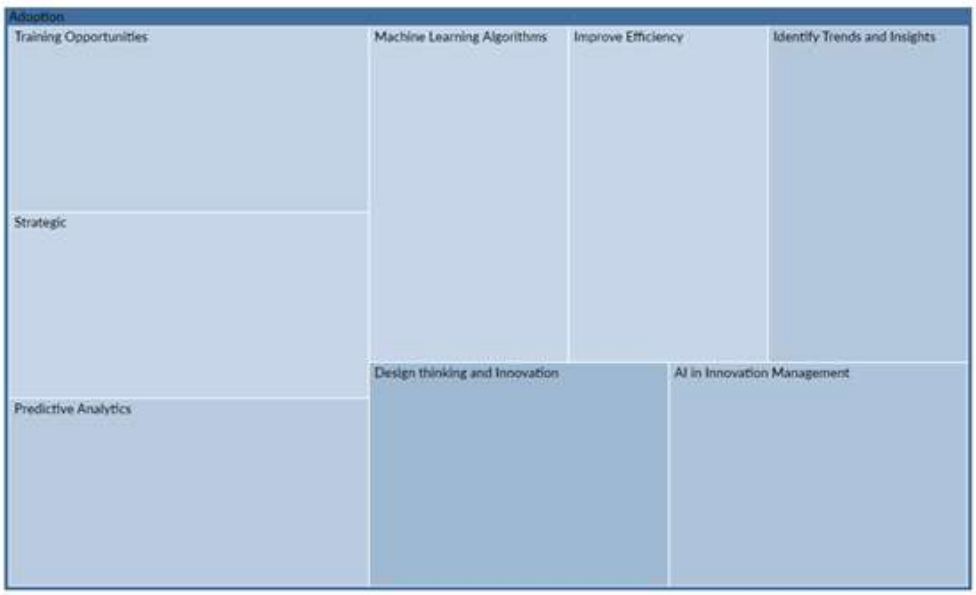
Qualitative Results

The interview study was conducted for eight members who held positions as employees or managers in different digital consultancies from the UK, such as CognitionX, Brainpool AI, Whoa, and The Data Lab. However, the data analysis of collected and transcribed interview data was performed using the NVivo 14 tool, and results were evaluated based on five different themes, such as adoption, user experience, challenges, limitations, and organizational effects of using AI in design thinking and innovation management of companies.

Adoption and User Experience of AI in Design Thinking and Innovation Management

Most of the interview responses mentioned that they adopted AI for different purposes, such as training opportunities, strategic management, design thinking processes, and innovation management. Some of the interviewees used machine learning algorithms to identify trends and insights in the adoption of AI into design thinking. For example, they adopted predictive analytics to enhance the efficiency of design thinking processes regarding market trends and customer insights (Figure 6). Thus, UK-based digital agencies adopted AI for efficient design thinking and innovation management through strategic vision capabilities.

Figure 6: Adoption of AI in Design Thinking and Innovation Management



Interviewees appreciated using AI tools due to their intuitive visualization and user-friendly dashboards, which would make the adoption of existing design thinking procedures easier. They also mentioned that AI-based tools provide data-driven predictive analytics to facilitate brainstorming sessions for new idea generation and innovation. Many of them reported that AI significantly automates tasks and performs data analysis for clear reporting of insights (Figure 7). Thus, it would allow the users to focus on creativity through customization to enhance efficiency in the design thinking and innovation processes.

Figure 7: User Experience of AI Tools in Design Thinking and Innovation Management

Challenges and Limitations of AI Systems in Design Thinking

The major challenges of AI tools included integration into existing design processes, and it is time-consuming to analyze the complex tasks according to the perspectives of some interviewees. Most of them raised issues with the use of AI tools in design thinking, and innovation management needed to be compatible, scalability, lack of flexibility, lack of real-time feedback, and customization (Figure 8). On the other hand, the challenges of data transfer issues, workflow disruptions, and data privacy and security were considered when using AI in design thinking procedures.

Figure 8: Challenges of Using AI Systems in Design Thinking Processes

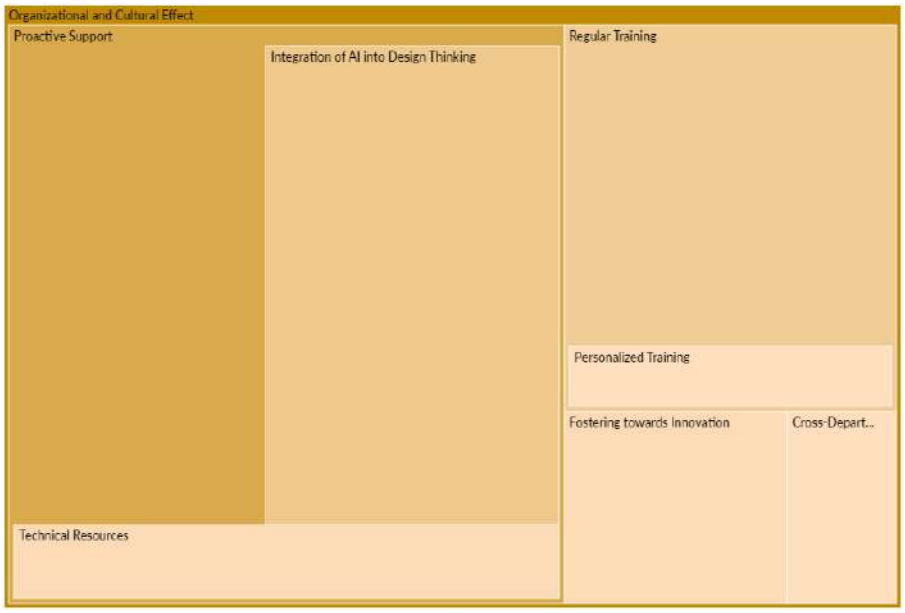
Some interviewees felt that one of AI's limitations was that there is always room for improvement since it involves advancements and evolutions. However, most of the interviewees opined that continuous learning and regular updates or additional training are required to be aware of AI tools for design thinking and innovation (Figure 9).

Figure 9: *Limitations of AI in Design Thinking and Innovation Management*

Organizational Effect of AI in Design Thinking and Innovation Management

Interviewees discussed receiving proactive support from their working digital consultancy companies from UK. However, organizational support was provided in terms of facilitating technical resources, integrating AI into design thinking, and regular training (Figure 10). Moreover, interviewees also mentioned that the organizations fostered an innovative culture by providing better collaboration among departments for idea generation.

Figure 10: Organizational Effect of AI in Design Thinking and Innovation



These qualitative data findings disclosed that AI-based design thinking and innovation provide various advantages for companies in terms of increased efficiency and encouraging innovation management. According to the experiences of employees and managers, these companies have effectively adopted AI tools for design thinking processes and innovation due to their user-friendly and intuitive visualization features. Notably, they faced challenges like integration, scalability, resource-intensive, and data privacy and security. To overcome these challenges, they provided proactive organization support, including the provision of regular training about integrating AI into existing design thinking processes and improved collaboration among team members.

DISCUSSION

The relationship between innovation management and AI-enhanced design thinking has gathered significant attention in recent decades. This discussion portion seeks to illuminate the integration of the above findings with previous literature, emphasizing both qualitative and quantitative data.

Summary of Quantitative Data Findings

The quantitative analysis shows several complex perspectives on the AI tools' efficacy in innovation management and design thinking. The research intended to investigate the stimulation of innovation management and AI-enhanced design thinking on IO ("innovation output"). However, the independent variables involved the ASE ("AI skillset and expertise"), LAI ("level of AI integration"), and the IAT ("implementation of AI technologies"). Therefore, this portion of the research discusses the outcomes regarding past literature, facilitating a comprehensive assessment of how AI impacts innovation management and design thinking.

AI Skillset and Expertise

Within an organization, this particular variable appeared as something other than a substantial analyst of innovative output. This outcome needs to be aligned with the research that consistently emphasizes human capital's significance in supporting AI for innovation. As per Fountaine et al. (2019), firms with an effective AI talent hub are well positioned to achieve the abilities of AI, converting into extreme innovation performance.

The research showed that AI expertise does not impact on innovation output of AI tools. On the contrary, Bughin et al. (2018), it is reported that organizations with skilled personnel and modern AI competencies obtained absolute innovation results. Therefore, skilled AI professionals' existence empowers companies to direct critical AI implementations, confirming robust utilization and implementation.

Level of AI Integration

This particular variable within firms also revealed an effective association with innovative output. Therefore, this reflects outcomes from previous studies that emphasized the significance of associating AI with several organizational roles to increase its potential. Thus, Davenport et al. (2018) stressed that organizations with deeper association of AI obtained better innovation results for increased collaboration and seamless information flow throughout departments.

The outcomes showed that higher AI integration levels resulted in more significant innovative output, indicating that a complete method of AI implementation is essential. However, this leverages the research of Haefner et al. (2021), where it is reported that organizations with associated AI systems throughout several roles, including R&D, operations, and marketing, implemented those with remote AI applications. Moreover, the association offers synergy, regulating more effective and creative innovation processes.

Implementation of AI Technologies

This variable was reported to have no significant impact on the innovative output. It is not associated with previous literature emphasizing the role of AI in shifting the processes of innovation. As an illustration, Wilson & Daugherty (2018) highlighted that AI tools, especially data analytics and machine learning, regulate innovation by promoting problem-solving competencies and empowering more advanced decision-making. Brynjolfsson & McAfee (2017) demonstrated that AI tools perform as a substance for innovation, permitting organizations to establish cutting-edge solutions and explore new paths.

In the research, firms that successfully adopted AI tools informed higher innovative output, leveraging the concept that AI seems pivotal factor of innovation. This outcome supports the study of Wang & Zhang (2020) that argued that organizations supporting AI tools in their processes of R&D perceived significant advancements in market performance and product development.

The research outcomes integrate well with past studies, strengthening the pivotal role of AI in promoting innovation management and design thinking. However, the positive effect of the level of AI integration on particular innovative output validates the presenting knowledge content, recommending that firms tend to acquire significant innovations obtained by concentrating on such areas. Therefore, previous research has shown that AI technologies regulate problem-solving, creativity, and efficiency (Agrawal et al., 2019). Thus, this study incorporates this by illustrating the particular ways in which AI integration to innovative output. Moreover, the alignment with the previous research emphasizes the effectiveness of the findings and their significance to the current assessment of the role of AI in innovation.

Summary of Qualitative Data Findings

The qualitative data analysis dependent on case studies and interviews, facilitates deeper overviews into how innovation management and AI-based design thinking are incorporated and perceived in practice.

- **Adoption and User Experience:** Interviewees often show that AI tools promote the user experience by offering intuitive interfaces and simplifying critical processes. Therefore, this insight reflects the outcomes of the literature, which observed that user-friendly interfaces and ease of use are essential factors in the AI tools' successful incorporation into design thinking (Lee & Park, 2021).
- **Limitations and Challenges:** The research also emphasizes some barriers despite the advantages, such as the integration of AI into design thinking methods, scalability, lack of flexibility, data privacy, data transfer, security, and time-consuming. However, this is relevant to the issues mentioned by the literature, which reported that while the equipment of AI facilitates substantial advantages, it also needs adjustment periods and significant training for users to become experts (Bughin et al., 2018).
- **Organizational Effect:** The qualitative data indicate that associating AI with innovation management and design thinking requires support from organizations. Respondents underscored the requirement for leveraging organizational support for training and integrating AI into existing design thinking procedures. This outcome is associated with the literature, which demonstrated that organizational support and culture possess a pivotal role in AI's successful implementation (Igna & Venturini, 2023).

The alignment of this research's findings with previous research highlights the transformative possibility of innovation management and AI-enhanced design thinking. In this research, the quantitative data leverage the idea that AI technology promotes data-driven decision-making, collaboration, and efficiency. Similarly, the qualitative data underscores the significance of user experience, the requirement for a leveraged organizational culture, and the adoption challenges.

CONCLUSION

This study provides an inclusive examination of the active association between innovation management, design thinking, and AI. Its mixed-method approach emphasizes key applications and findings.

The AI's stimulation on engineering-based innovation competency is determined. However, the study shows that effective AI association in the process of innovation results in advanced innovative outcomes. Therefore, AI technology empowers engineers to iterate, prototype, and experiment more effectively, thus stimulating a culture of continuous innovation and improvement. The research clarifies the substantial effect of AI on design thinking. AI promotes efficiency, increases collaboration within teams, and provides data-operated decision-making. Organizations lead by supporting updated analytics and perceptions to regulate more effective and informed innovative strategies.

The interaction between AI technology and design thinking has a synergistic impact that strengthens the innovation process. AI facilitates the data-driven perceptions and computational power required to balance design thinking's human-centered approach. Thus, this combination results in a deeper assessment and innovative solution to user preferences and requirements. The level of AI skills and expertise, adoption, and integration in AI tools are essential components stimulating the success of innovative solutions. However, research reveals that firms with higher degrees of AI-skilled workforce and integration in AI equipment obtain better innovation results. Such outcomes emphasize the significance of investing in AI development and training to connect the comprehensive possibility of AI in innovation management and design thinking.

The quantitative research facilitates a comprehensive understanding of AI technology's efficacy in design thinking and innovation management. It evaluates several metrics to ascertain the AI's robustness in promoting such processes. However, this analysis accounts for diverse independent variables such as IAT ("implementation of AI technologies"), LAI ("level of AI integration"), and ASE ("AI skillset and expertise"). Therefore, such variables seem essential in assessing how AI participates in IO ("innovation output").

The extent to which AI tools are applied also plays a pivotal role. The study reveals that the complete incorporation of AI technologies across several levels of design thinking procedures results in more innovative and effective solutions. However, this outcome indicates that a complete method of AI incorporation led to significant advantages in the context of innovation management. Therefore, the research shows that the AI integration level within a firm substantially affects its innovation

competency. Firms that are completely associated with AI in their respective systems seek to display higher innovation results. This association permits for seamless partnership between human creativity and AI systems, stimulating a more inclusive and innovative culture. Thus, this study emphasizes the significance of possessing a skilled workforce capable of AI technologies. Therefore, it obtains that firms with an effective ASE stand efficiently incorporated to support AI for creative and innovative purposes. Moreover, development and training in AI are essential in increasing its possibility in the process of innovation.

Apart from that, the qualitative research findings balance the quantitative outcomes by offering contextual depth. Using case studies and interviews, the study examines the practical implications of AI-advanced innovation management and design thinking. Therefore, such perspectives show successes and practical barriers, facilitating a subtle assessment of AI's impact.

AI-developed design thinking substantially promotes efficiency by offering frequent data analysis and systematizing repetitive tasks. This permits certain human designers to concentrate on more strategic and creative aspects of the system. Furthermore, AI leverages data-operated decision-making, allowing firms to make more advanced choices dependent on complete data perceptions. However, the association of AI stimulates a collaborative environment by facilitating technologies that provide teamwork and communication. Moreover, the systems of AI lead to construct a distance or gap between streamlined workflows and departments, resulting in more innovative and cohesive remedies. Thus, the study emphasizes the requirement for a leveraging organizational environment that holds AI and fosters continuous adaptation and learning.

The research illustrates that AI-developed innovation management and design thinking facilitate significant advantages, consisting of enhanced collaboration, developed decision-making, and increased efficiency. The study also underscores certain challenges or barriers involving the requirement for a leveraging organizational environment and the basic learning curve integrated with AI association. Covering such barriers seems essential for firms to completely realize the AI benefits in regulating innovation. Apart from that, the research also suggests a strategic roadmap development, AI infrastructure investment, and training programs for AI integration. Finally, such measures lead to assist firms in fully obtaining AI advantages in innovation management and design thinking.

Implications for Practitioners

For practitioners, the research implications of this study are multifaceted:

- Balancing AI and Human Inputs: It is essential to maintain AI inputs with intuition and human creativity to ignore over-dependency on technology while AI technology facilitates significant advantages.
- Strategic Implementation: Organizations need to adopt AI technologies strategically in their innovation processes and design, confirming adequate support and training for users.
- Investing in AI Tools: Organizations need to prioritize the incorporation of AI technologies and tools that align with their respective innovative objectives. Therefore, this includes investing in AI-driven design tools, advanced machine learning algorithms, and data analytics platforms.
- Stimulating a Leveraging Culture: Directing a culture that embraces experimentation and innovation is pivotal for AI's successful association with innovation management and design thinking.
- Building AI Expertise: Building a skilled workforce in AI seems essential. Organizations need to invest in several training programs and attract top talent in AI to promote their abilities. Therefore, this includes stimulating a continuous learning culture and staying advanced with modern AI improvements.
- Stimulating AI Integration: Organizations need to endeavour for a high AI integration level throughout diverse functions to increase the AI's advantages. Therefore, this needs to foster cross-functional collaboration and break down silos, confirming that AI overviews are supported across the firms.

This research illustrates that innovation management and AI-enhanced design thinking facilitate significant benefits in the case of efficiency, data-regulated decision-making, and collaboration. Such findings associate closely with the present study, emphasizing the transformative AI potentiality in these segments. Hence, obstacles consisting of the requirement for a leveraging organizational culture and primary learning curve need to be covered to comprehensively observe the AI integration's advantages. Therefore, the applications for academic practitioners and the paths for future investigation emphasize the significance of strategic implementation and continued exploration of AI in innovation practices and design. The research ensures that the AI expertise and skillset, the AI integration level, and AI technologies' implementation significantly foster innovative output. Thus, such outcomes are relevant to previous studies, highlighting the significance of AI in the context of driving innovation. Organizations can promote their innovation management and design thinking processes, ultimately obtaining higher innovative output levels by concentrating on such major areas. Moreover, further study can again investigate the particular mechanisms by which AI affects innovation, facilitating deeper perceptions of such multifaceted associations.

For industry leaders and academic practitioners, the research underscores the significance of strategic AI application and continuous examination of its effect on several innovation practices. However, future research needs to explore the particular mechanisms by which AI impacts innovation, facilitating deeper perceptions of such multifaceted associations. Organizations promote their design thinking and innovation management processes, finally obtaining higher degrees of innovative results by concentrating on major segments, including technology implementation, integration levels, and AI expertise.

The study recommends diverse approaches for future research. Such avenues involve examining the particular mechanisms or tools by which AI fosters innovation, evaluating the AI association's long-term effects on organizational performance, and determining the appropriate practices for AI incorporation in several sectors. Future research tends to explore AI's ethical applications in innovation, confirming that AI-developed design thinking is associated with wider societal values. The research again emphasizes the transformative possibility and competency of AI in innovation management and design thinking. Organizations can disclose new scopes for competitive advantage, efficiency, and creativity, making the path for sustained growth and innovation by strategically associating AI with such processes. However, the study facilitates a complete infrastructure for supporting and assessing AI in innovation management and design thinking by meeting such factors. Moreover, firms that hold such perspectives can promote their innovation quality, maintain a competitive culture, and obtain a higher degree of innovative results in a highly AI-regulated world.

Recommendations

Organizations need to prioritize several training programs to increase the AI expertise and skills of employees. These programs involve certification programs, online courses, and workshops concentrated on AI technologies and tools appropriate to innovation and design thinking. The research study directed to stimulate a continuous learning culture where employees are encouraged to remain advanced with the modern developments in AI and its implementation in their segments (Ghosh, 2019).

The organization should also establish a strategic roadmap or planning for AI implementation throughout all stages of the process of design thinking. Therefore, this roadmap contains implementation, testing, prototyping, and ideation. The firms need to stimulate collaboration between diverse departments (for example, production, marketing, R&D) to confirm that AI association is integrated and complete with entire organizational objectives. Apart from that, the firms need to incorporate AI-powered analytics to collect and evaluate the information from several sources. This assists in possible innovation opportunities, user preferences, and determining trends (Fountaine et al., 2019).

The organizations incorporate AI-regulated predictive approaches to estimate consumer behaviour and market trends, helping to improve innovative services and products. Additionally, the firms should spend on AI-developed collaborative technologies that provide teamwork and communication throughout diverse departments and teams. Such technologies contain AI-driven chatbots, project management software, and virtual or digital collaboration platforms. It suggests that firms should organize brainstorming sessions and workshops that encourage employees from several departments to associate with the projects of AI-developed innovation (Descour et al., 2021).

The firms should also cultivate an environment that holds AI and fosters innovation and experimentation. This culture or environment covers any struggle to evolve, transparent interaction of AI advantages, and leadership support. Additionally, they need to implement recognition and reward programs for respective employees who participate in AI-developed innovation projects, stimulating a sense of motivation and ownership. Initiate several pilot programs to evaluate AI technologies and tools to a smaller degree. Therefore, this can assist in ascertaining possible obstacles and refining several systems before full-scale application. Moreover, developing mentorship programs can also be effective where experienced and skilled professionals lead to support and guide employees new to such AI tools (Davenport et al., 2020).

Future Research Directions

In the study, the undertaken journey opens several paths to future explorations. However, the leadership dynamics in the context of AI-developed innovation management and design thinking warrant continuous exploration (Obuobisa-Darko, 2015; Li et al., 2019). However, further research can examine the particular mechanisms through which artificial intelligence interacts with several leadership practices, facilitating deeper overviews of the changing organizational leadership landscape (Oprean-Stan et al., 2020). The study implies that the association between design thinking, innovation management, and AI seems dynamic and leads to change. It is assumed that AI tools will continue to change, facilitating more sophisticated capabilities and tools for innovation management and design thinking. Improvements in areas including machine learning, computer vision, and natural language processing seek to empower more intuitive and advanced AI implementations. However, the future systems of AI also promote human creativity by developing new solutions, designs, and ideas, further associating AI with design thinking's creative aspects (Ghosh, 2019).

The association of artificial intelligence with the IoT lead to more responsive and smarter services and products. This process of design thinking will effectively adopt data from IoT technologies to conduct more adaptive and personalized

solutions. VR ("virtual reality") and AR ("augmented reality") tools integrated with AI will facilitate immersive cultures for testing and prototyping innovative ideas, permitting for more robust user feedback and visualization (Iansiti & Lakhani, 2020; Lee, 2019; Landers et al., 2017; Koohang et al., 2017). Furthermore, AI systems are going to be effective at assessing context, collaborating with humans, and facilitating relevant assistance. However, this will provide a more seamless association of AI into the processes of design thinking, promoting the human teams' competencies (Pfau & Rimpp, 2021). Moreover, future AI technologies will be developed with a larger concentration on human-centric design and user experience principles. It is making them more accessible and intuitive to a wider user range (Kumar et al., 2019).

The conclusion, in essence, caters as a reflection on the current leadership state, innovation management, AI, and design thinking in the UK manufacturing sector and also to future horizons' contemplation. The research further challenges organizations as well as leaders to leverage technology, navigate complexities, and embrace change for operational efficiency and as equipment for influencing innovation and human creativity. Therefore, the study's conclusion is a call to perform for several leaders and not an endpoint to embark on ethical leadership and a continuous adaptation journey in the changing environment of technology integration and manufacturing. The research also represents a demand for action, recommending leaders embrace transformation and actively construct the future by encouraging a creative culture, associating technology responsibly, and resulting in several ethical considerations. Finally, the research culmination seems a conclusion and an invitation to force on a landscape where leadership directs organizations towards potential uncharted territories powered by AI and fueled by innovation.

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APPENDICES

Appendix A: Quantitative Data Findings

Table A1: *KMO and Bartlett's Test – Factor Analysis*

KMO and Bartlett's Test		
Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		0.900
Bartlett's Test of Sphericity	Approx. Chi-Square	20253.198
	df	496
	Sig.	0.000

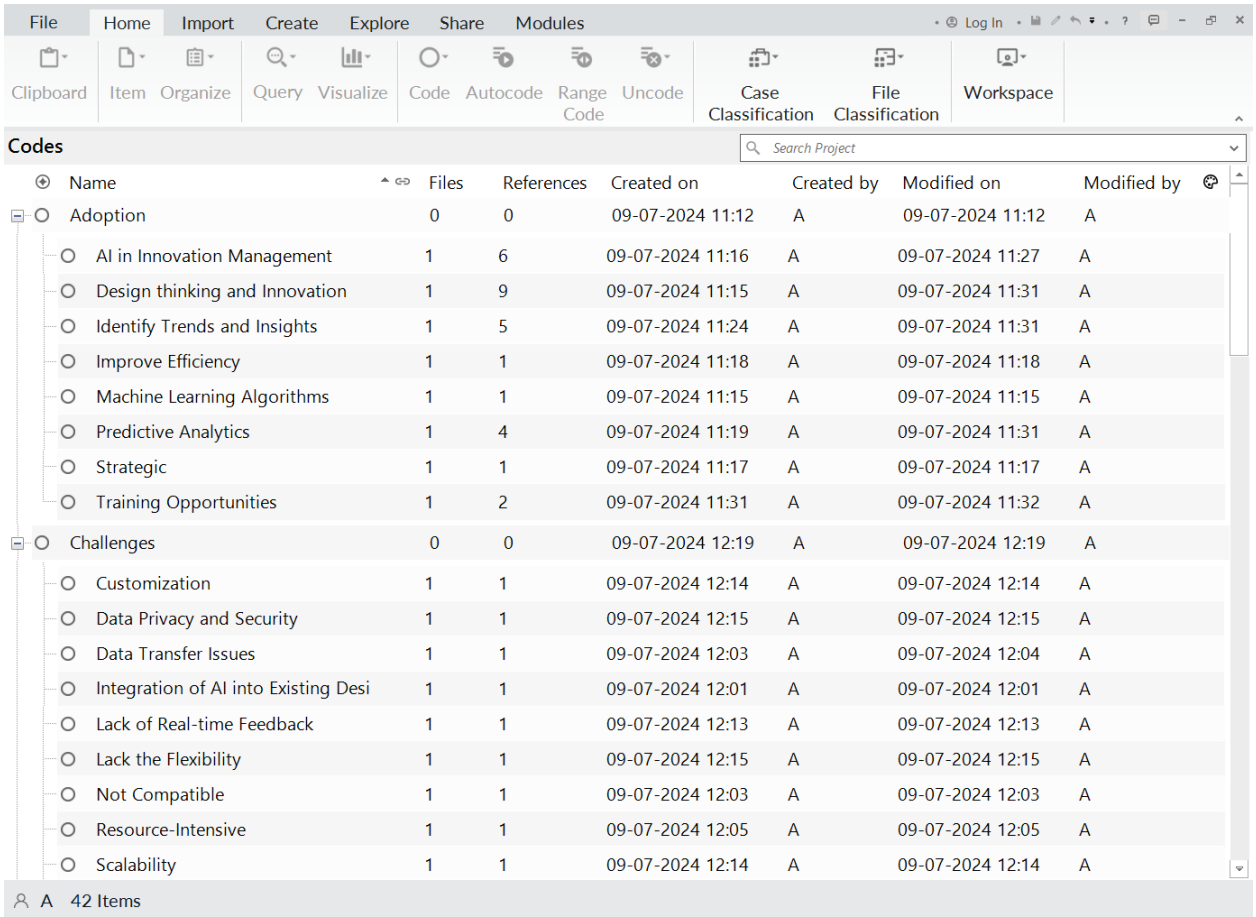
Table A2: *Rotated Component Matrix – Factor Analysis*

Rotated Component Matrix[a]						
	Component					
	1	2	3	4	5	6
IAT1	0.478	0.656	0.123	0.196		0.332
IAT2	0.450	0.210	0.181			0.725
IAT3	0.734	0.156	0.145		0.117	0.513
IAT4	0.378	0.530	0.161	0.165	0.142	0.564
IAT5	0.433	0.485	0.124		0.172	0.602
IAT6	0.904	0.202		0.189		0.133
IAT7	0.903	0.206		0.171		0.136
IAT8	0.701	0.420		0.124		0.108
LAI1	0.563	0.662		0.201		0.213
LAI2	0.912	0.180	0.125			0.169
LAI3	0.874	0.230	0.126		0.118	0.171
LAI4	0.681	0.328			0.158	-0.130
LAI5	0.581	0.657	0.123		0.113	0.238
LAI6	0.910	0.242	0.103	0.104		0.132
LAI7	0.881	0.262	0.124			0.105
LAI8	0.547	0.750	0.134	0.122		0.203
ASE1	0.406	0.501	-0.164			-0.281
ASE2	0.219	0.865				
ASE3	0.192	0.888				
ASE4	0.200	0.813		-0.101		0.177
ASE5	0.570	0.349	0.432			0.301
ASE6	0.560	0.392	0.458			0.298
ASE7	0.145		0.801	0.471		
ASE8	0.144		0.767	0.542		
IO1	0.106		0.173	0.829	0.421	
IO2	0.104		0.172	0.845	0.360	
IO3	0.123		0.232	0.823	0.384	
IO4	0.103		0.823		0.463	0.122
IO5			0.184	0.368	0.880	
IO6			0.183	0.385	0.844	
IO7			0.238	0.259	0.876	
IO8	0.115		0.839	0.146	0.369	0.120

Note. a. Rotation converged in 8 iterations using Principal Component Analysis

Appendix B: Qualitative Data Findings

Figure B1: *Codes for Interview Data*



Codes

Name	Files	References	Created on	Created by	Modified on	Modified by
Adoption	0	0	09-07-2024 11:12	A	09-07-2024 11:12	A
AI in Innovation Management	1	6	09-07-2024 11:16	A	09-07-2024 11:27	A
Design thinking and Innovation	1	9	09-07-2024 11:15	A	09-07-2024 11:31	A
Identify Trends and Insights	1	5	09-07-2024 11:24	A	09-07-2024 11:31	A
Improve Efficiency	1	1	09-07-2024 11:18	A	09-07-2024 11:18	A
Machine Learning Algorithms	1	1	09-07-2024 11:15	A	09-07-2024 11:15	A
Predictive Analytics	1	4	09-07-2024 11:19	A	09-07-2024 11:31	A
Strategic	1	1	09-07-2024 11:17	A	09-07-2024 11:17	A
Training Opportunities	1	2	09-07-2024 11:31	A	09-07-2024 11:32	A
Challenges	0	0	09-07-2024 12:19	A	09-07-2024 12:19	A
Customization	1	1	09-07-2024 12:14	A	09-07-2024 12:14	A
Data Privacy and Security	1	1	09-07-2024 12:15	A	09-07-2024 12:15	A
Data Transfer Issues	1	1	09-07-2024 12:03	A	09-07-2024 12:04	A
Integration of AI into Existing Desi	1	1	09-07-2024 12:01	A	09-07-2024 12:01	A
Lack of Real-time Feedback	1	1	09-07-2024 12:13	A	09-07-2024 12:13	A
Lack the Flexibility	1	1	09-07-2024 12:15	A	09-07-2024 12:15	A
Not Compatible	1	1	09-07-2024 12:03	A	09-07-2024 12:03	A
Resource-Intensive	1	1	09-07-2024 12:05	A	09-07-2024 12:05	A
Scalability	1	1	09-07-2024 12:14	A	09-07-2024 12:14	A

Figure B2: *Themes for Interview Data*